

PUBLISHED: 25-AUG-2017
2014.0 RANGE ROVER (LG), 303-01

ENGINE - V8 S/C 5.0L PETROL

CRANKSHAFT PULLEY (G1851000)

REMOVAL AND INSTALLATION

12.21.01	PULLEY - CRANKSHAFT - RENEW	5000 CC, AJ V8, SUPERCHARGED	2.3	USED WITHINS	
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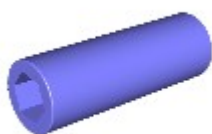
SPECIAL TOOL(S)



E115256

303-1437

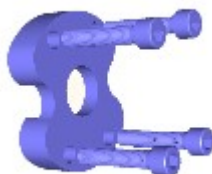
Crankshaft Damper Remover
/Installer



E115257

303-1438

Crankshaft Damper Bolt
Remover/Installer



E115258

303-1439

Crankshaft Damper Removal
Plate



E115259

303-1440

Crankshaft Damper Removal
/Installation Stud



E115260

303-1441

Crankshaft Damper Remover
/Installer Body



E115266

303-1448

Locking Tool



E152576

JLR-303-1304

Locking Tool

PART(S)

STEP	PART NAME	QUANTITY
Installation Step 11	Crankshaft pulley bolt	1

REMOVAL

NOTES:

- This procedure contains some variation in the illustrations depending on the vehicle specification, but the essential information is always correct.
- This procedure contains illustrations showing certain components removed to provide extra clarity.

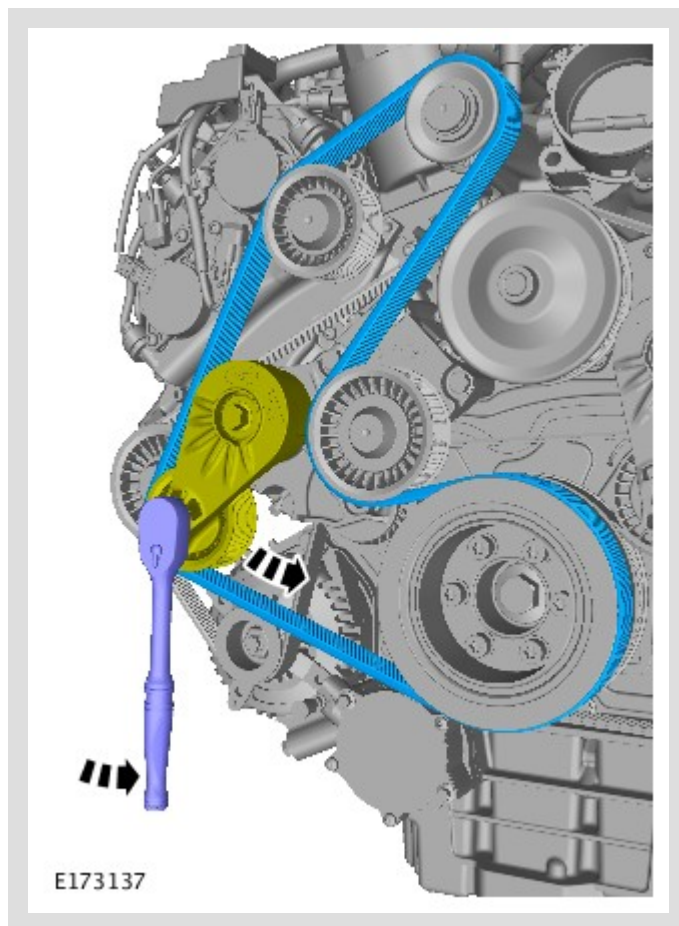
1. Disconnect the startup battery ground cable.
Refer to: Specifications (414-01 Battery, Mounting and Cables, Specifications).

2. Raise and support the vehicle on a suitable 2 post lift.
Refer to: Lifting (100-02 Jacking and Lifting, Description and Operation).

3. Remove the cooling fan.
Refer to: Cooling Fan Motor and Shroud (303-03 Engine Cooling - V6 S/C 3.0L Petrol, Removal and Installation).

4. Remove the starter motor.
Refer to: Starter Motor (303-06 Starting System - V6 S/C 3.0L Petrol, Removal and Installation).

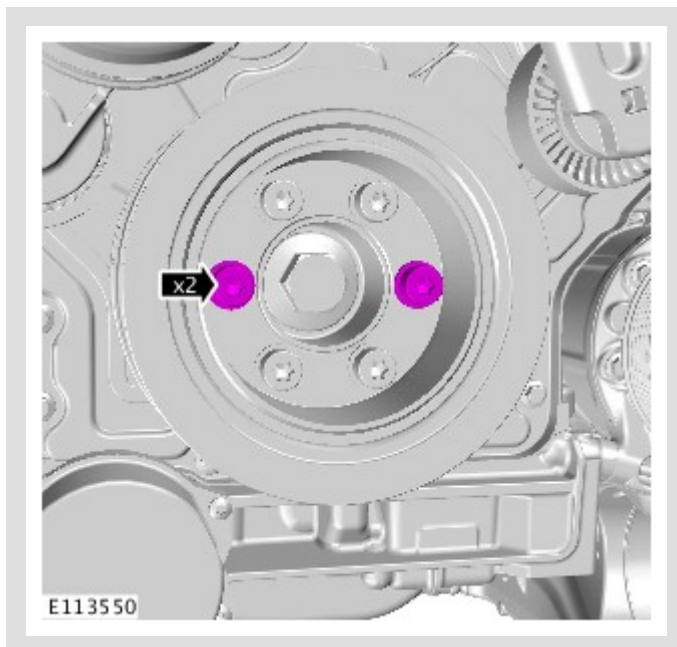
5.



Remove the supercharger drive belt.

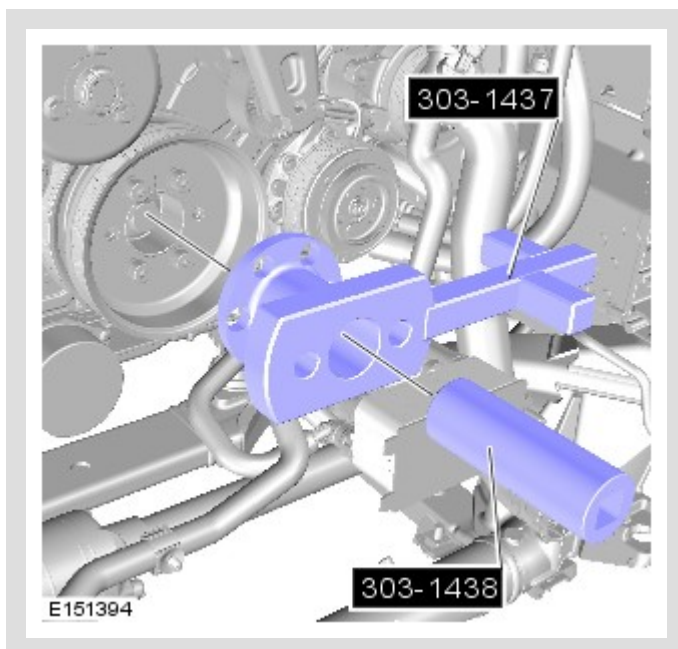
- Install the special tool.
- Vehicles without Auto stop/start system.
Special Tool(s): [303-1448](#)
- Vehicles with Auto stop/start system.
Special Tool(s): [JLR-303-1304](#)

8.

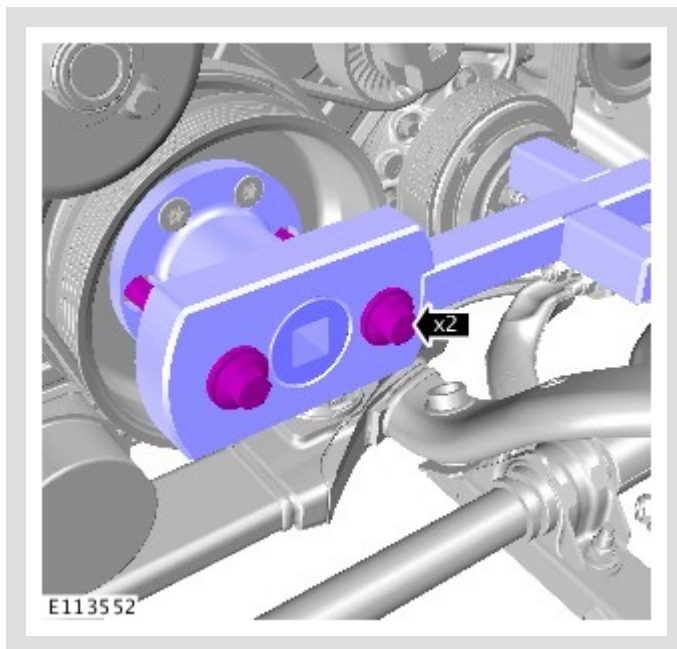


Remove the bolts.

9.



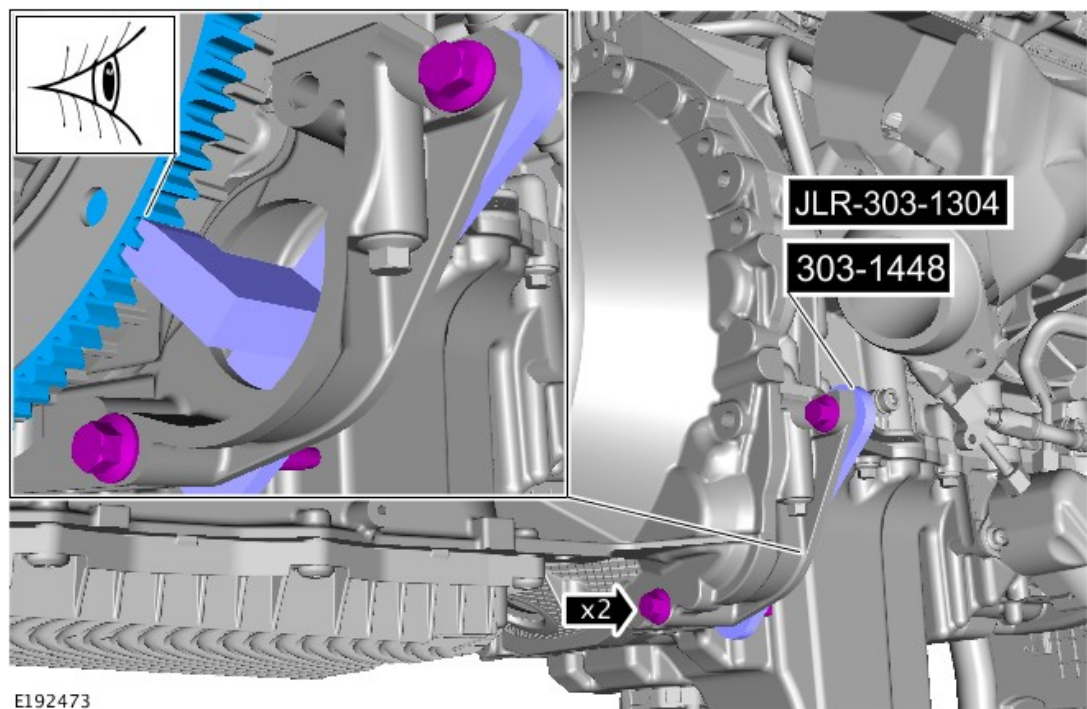
- Install the special tools.
- *Special Tool(s):* [303-1437](#)
- *Special Tool(s):* [303-1438](#)



Install the bolts.

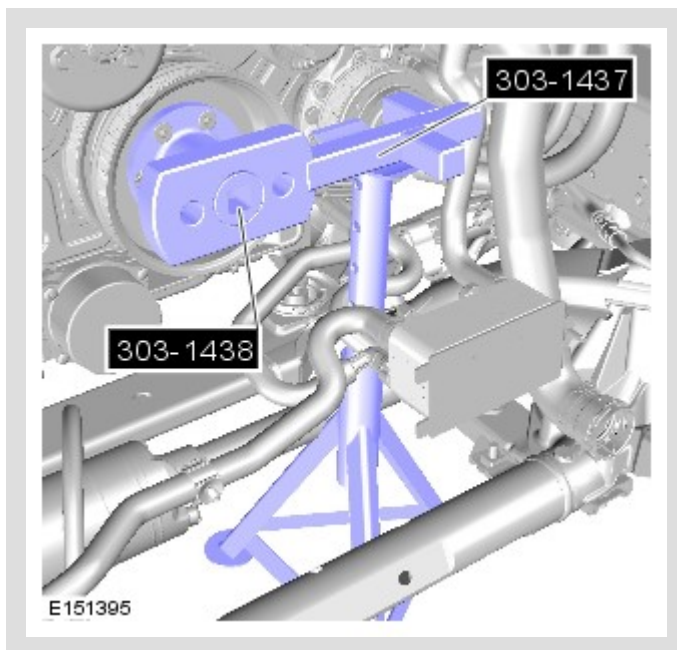
Torque: 25 Nm

11.



- Remove the special tool.
- Vehicles without Auto stop/start system.
Special Tool(s): [303-1448](#)
- Vehicles with Auto stop/start system.
Special Tool(s): [JLR-303-1304](#)

12.

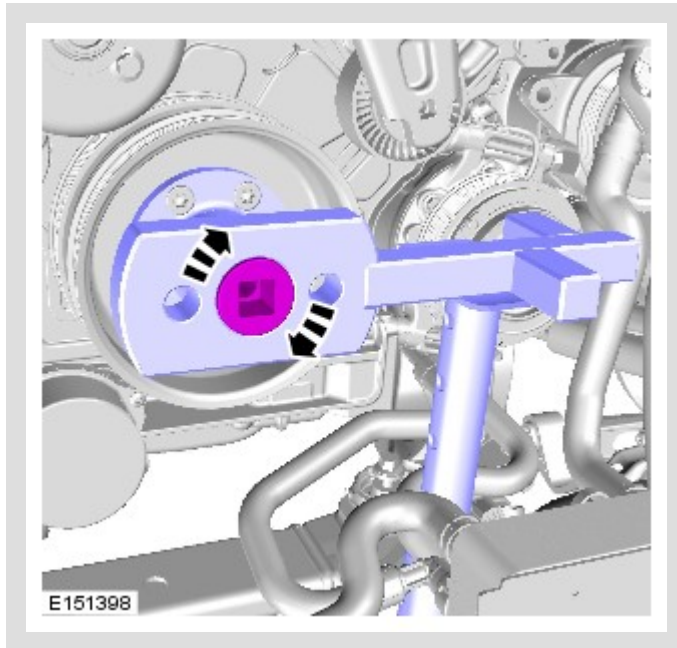


- Using a suitable vehicle stand, support the special tools.
- *Special Tool(s):* [303-1437](#)
- *Special Tool(s):* [303-1438](#)

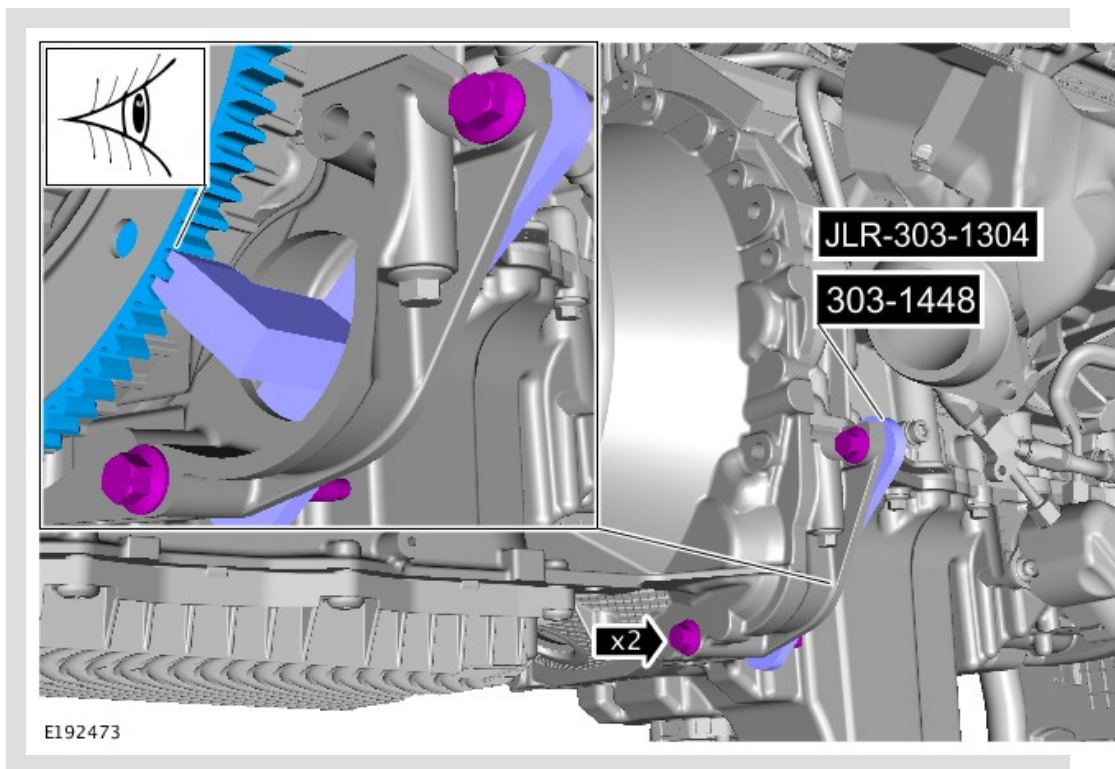
13.

 NOTE:

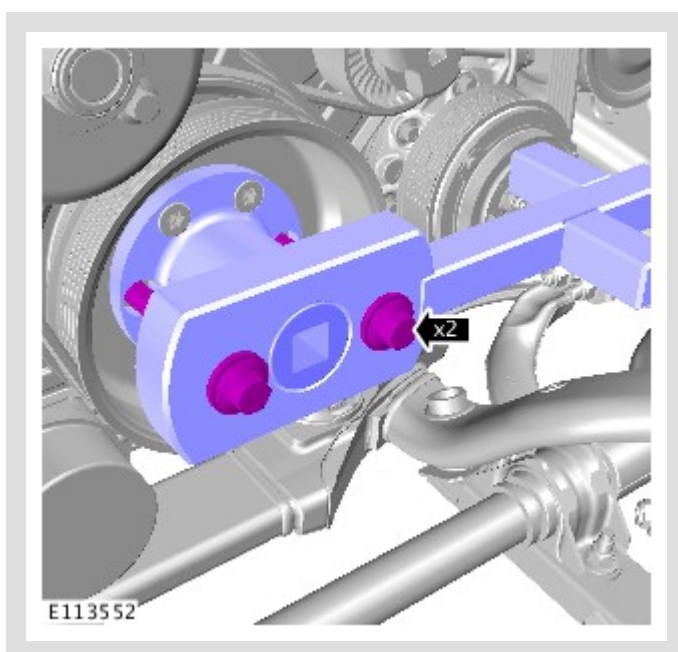
The crankshaft pulley bolt will be very tight.



Remove the crankshaft pulley bolt.

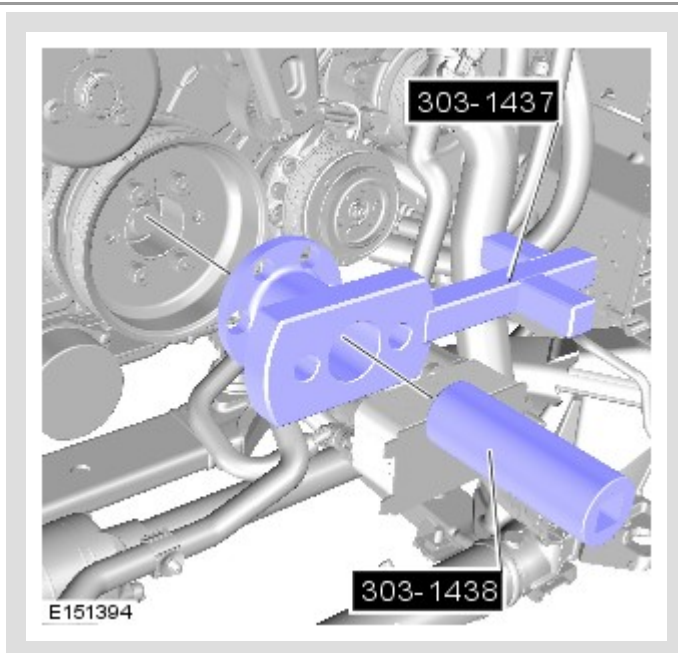


- Install the special tool.
- Vehicles without Auto stop/start system.
Special Tool(s): [303-1448](#)
- Vehicles with Auto stop/start system.
Special Tool(s): [JLR-303-1304](#)



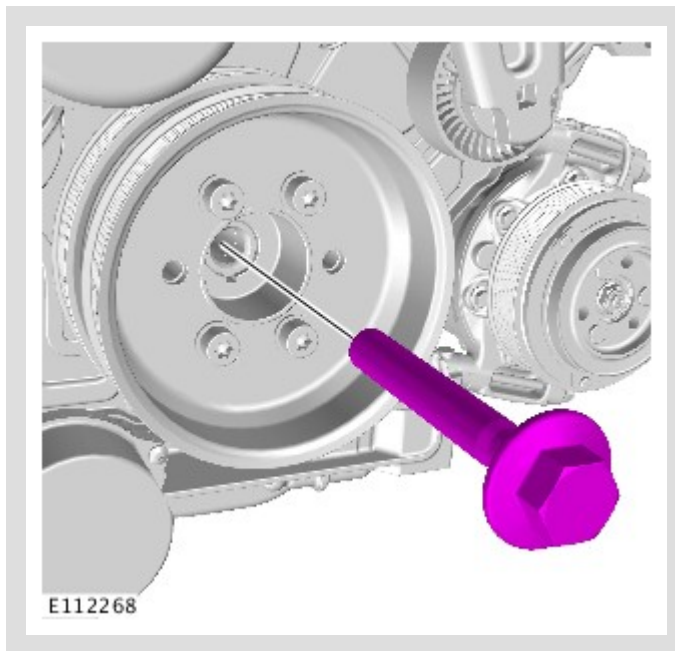
- Remove the two bolts.

16.



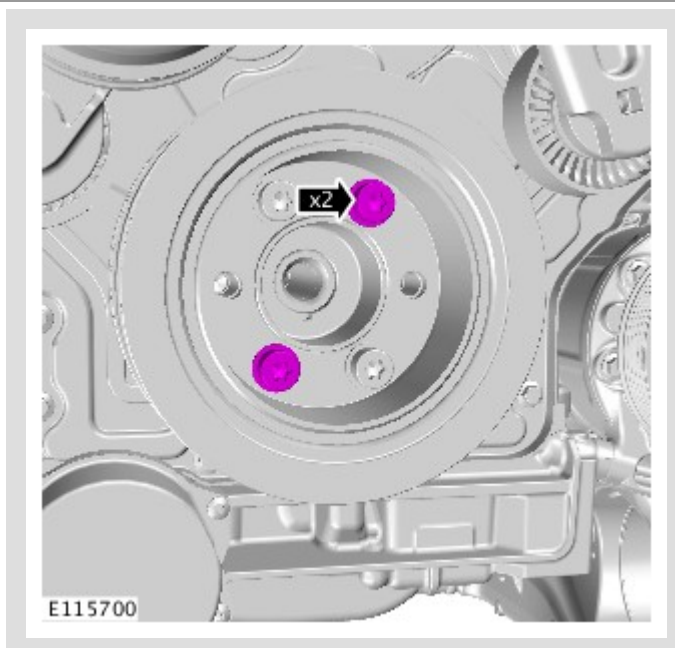
- Remove the special tools.
- Special Tool(s):* [303-1437](#)

17.

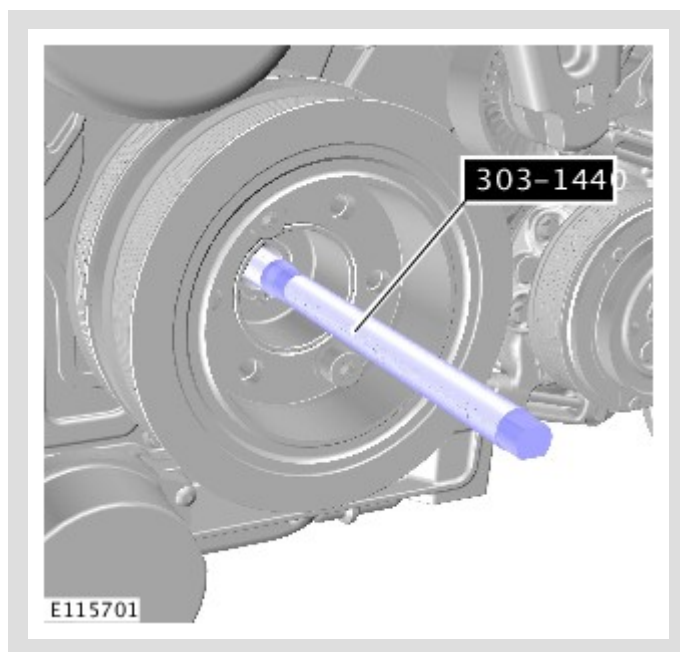


Remove and discard the bolt.

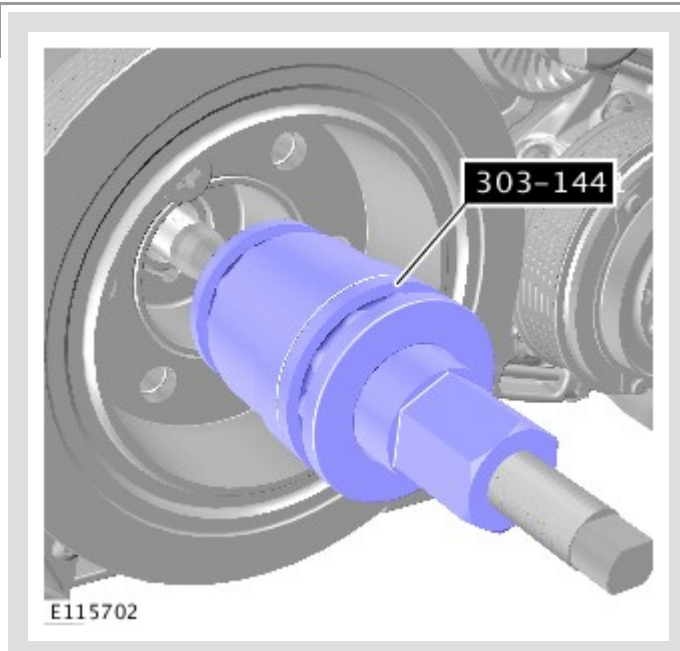
18.



Remove the bolts.

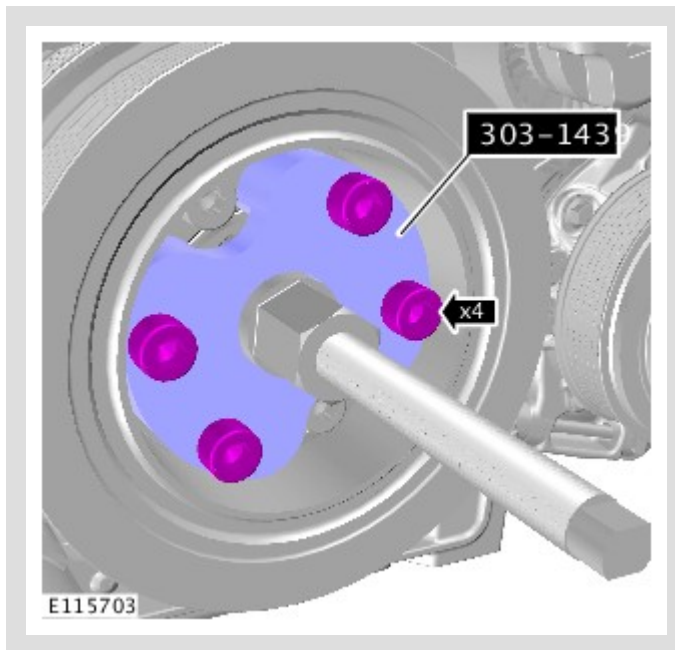


- Install the special tool.
- *Special Tool(s):* 303-1440



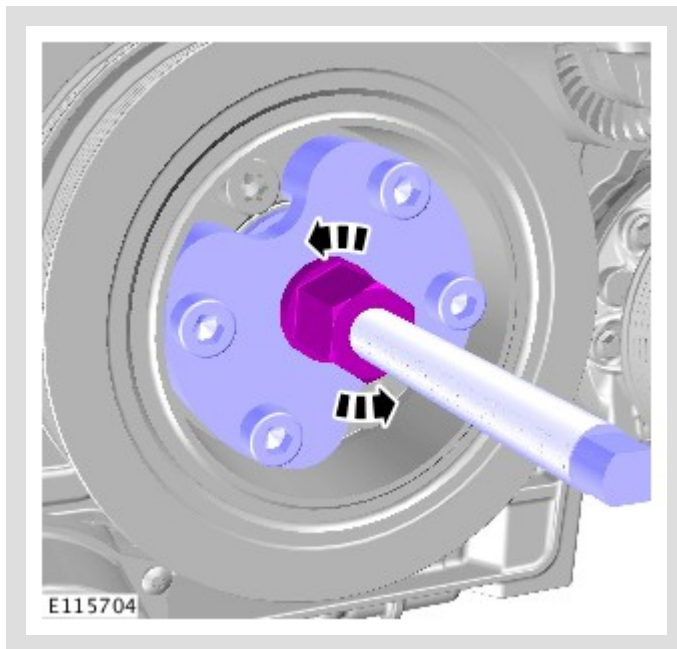
- Install the special tool.
- *Special Tool(s):* [303-1441](#)

21.



- Install the special tool.
- *Special Tool(s):* [303-1439](#)

22.

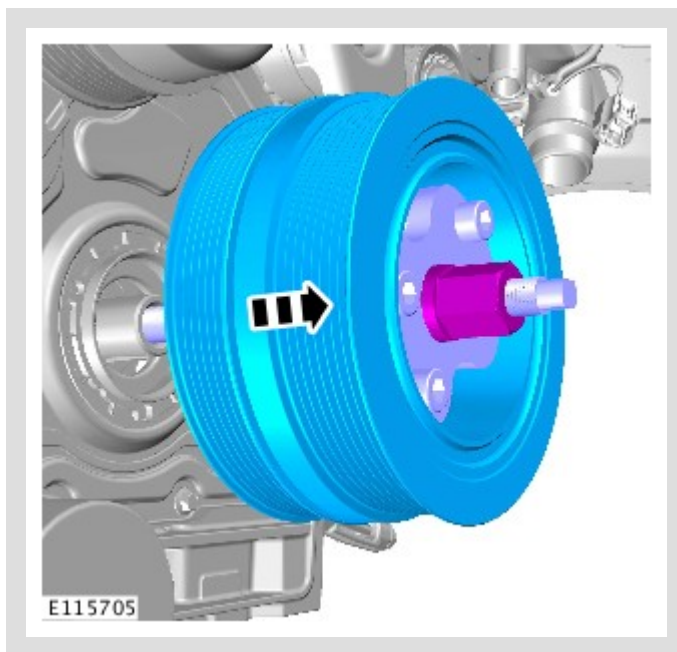


Rotate the special tool nut to remove the crankshaft pulley.

23.

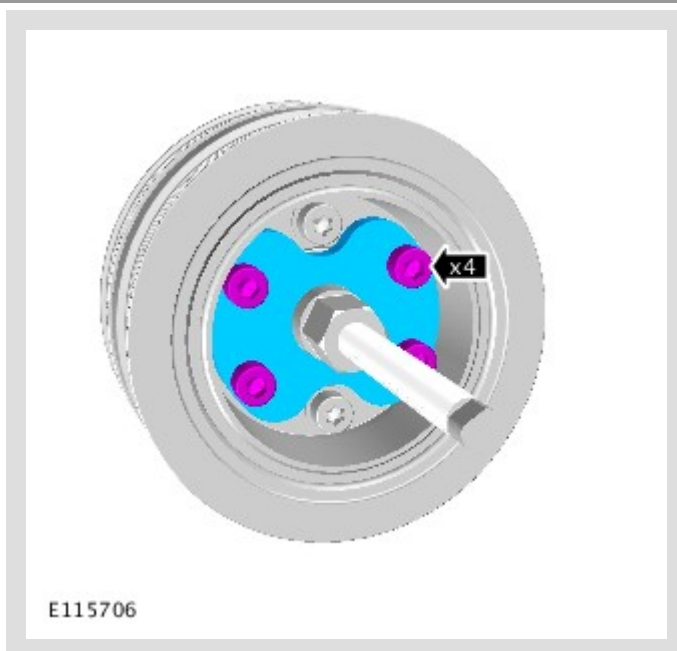
 NOTE:

Make sure to clean the threads in the end of the crankshaft and that the crank nose is free of any foreign materials.

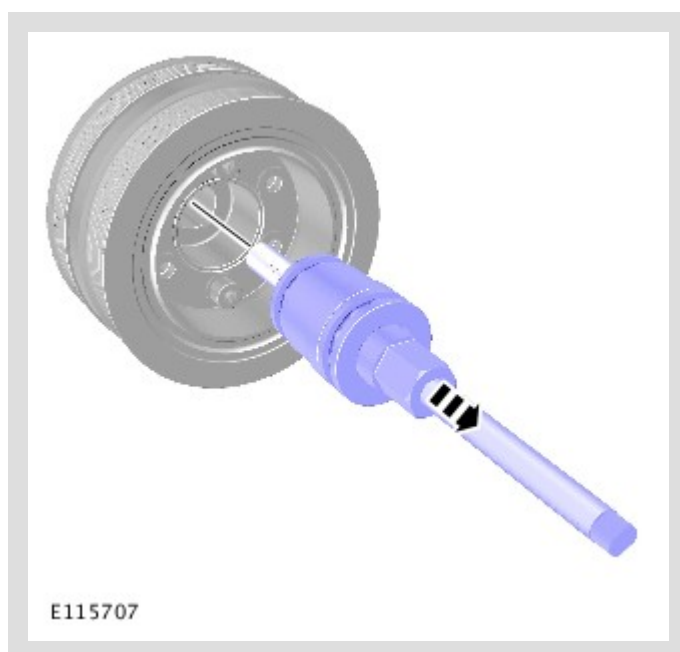


- Remove the crankshaft pulley.

24.



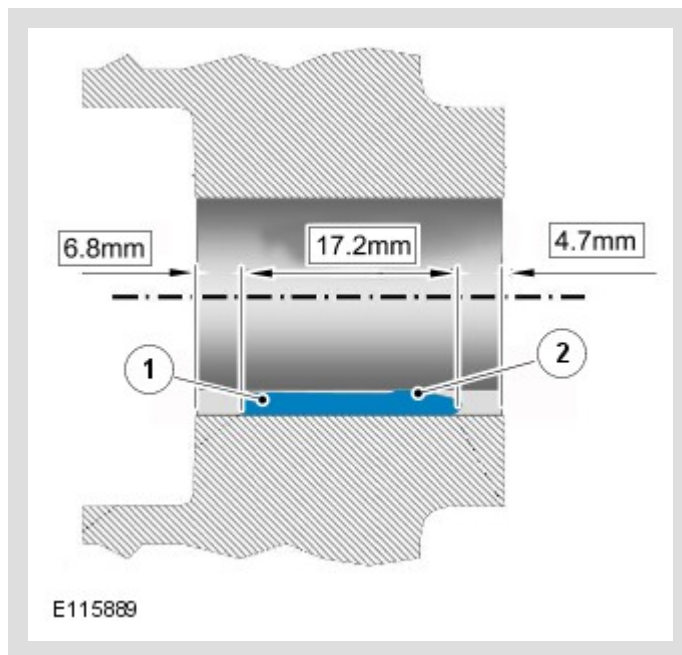
Remove the special tool.



Remove the special tool.

INSTALLATION

1.

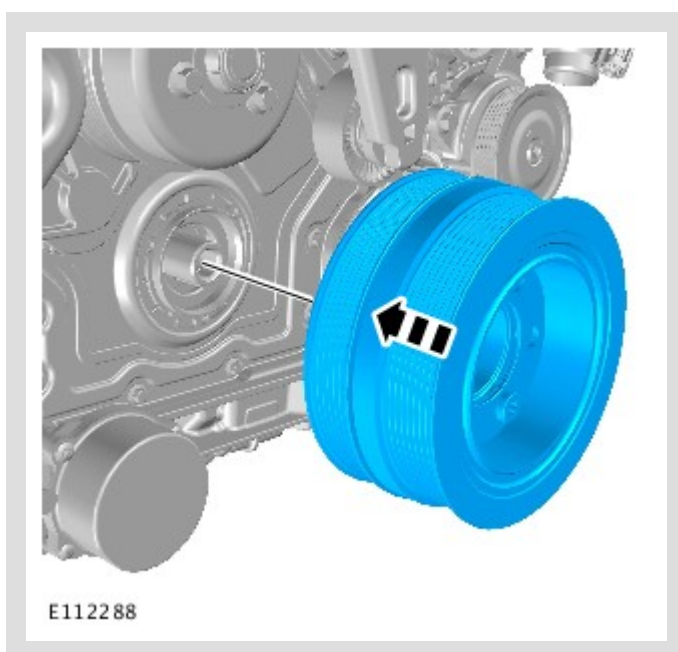


- Apply RTV sealant to the crankshaft pulley keyway.
- Make sure that the RTV sealant is applied in a 2mm diameter bead.
- Make sure that when the RTV sealant is applied that the RTV sealant is level with the top of the keyway.

2.

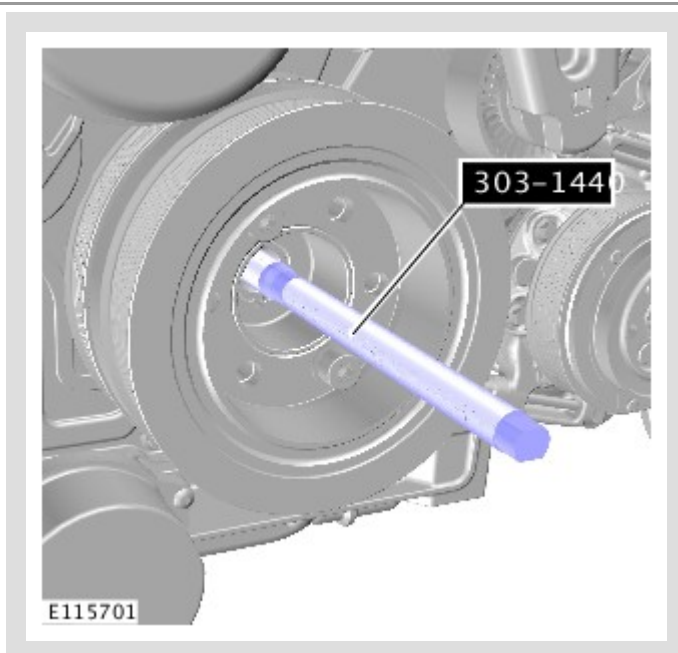
 NOTE:

Make sure to clean the threads in the end of the crankshaft and that the crank nose is free of any foreign materials.

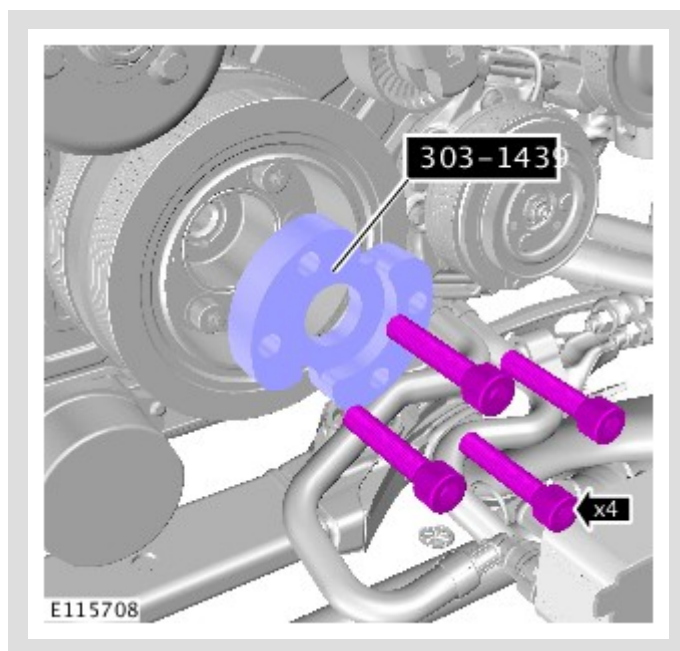


- Install the crankshaft pulley.

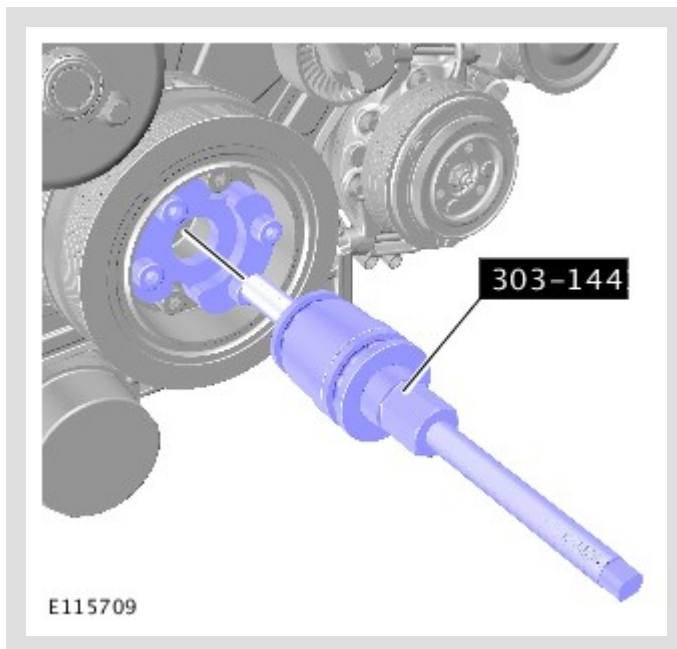
3.



- Install the special tool.
- *Special Tool(s):* [303-1440](#)



- Install the special tool.
- *Special Tool(s):* [303-1439](#)
- *Torque:* **25 Nm**

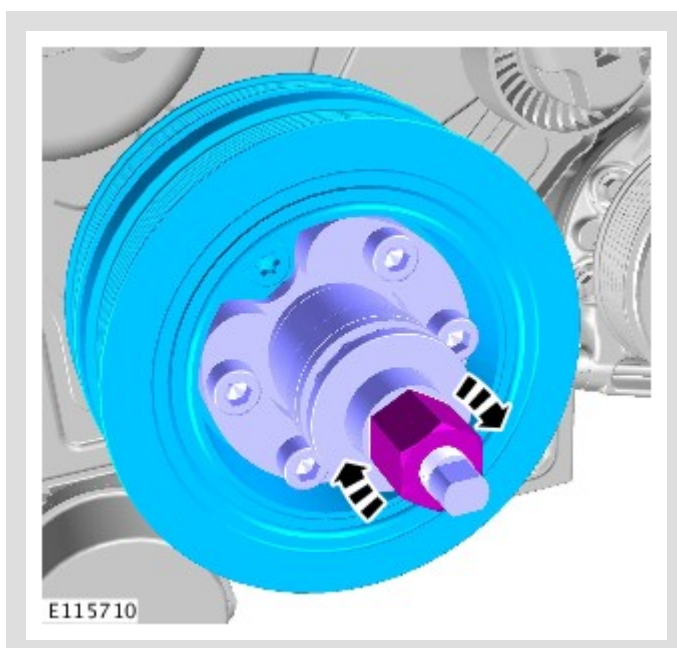


- Install the special tool.
- *Special Tool(s):* [303-1441](#)

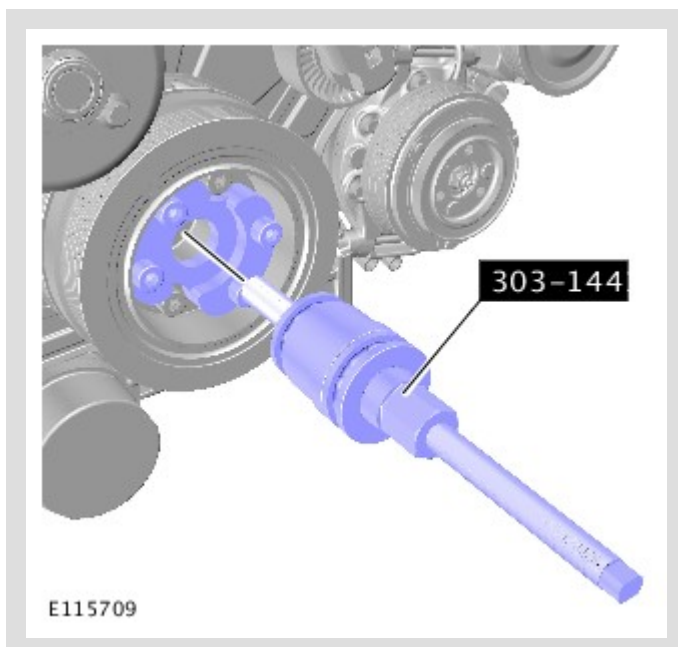
6.

⚠ CAUTION:

Rotate the crankshaft pulley installation tool until the pulley is fully located, do not over tighten. Failure to do this may result in damage to the components.

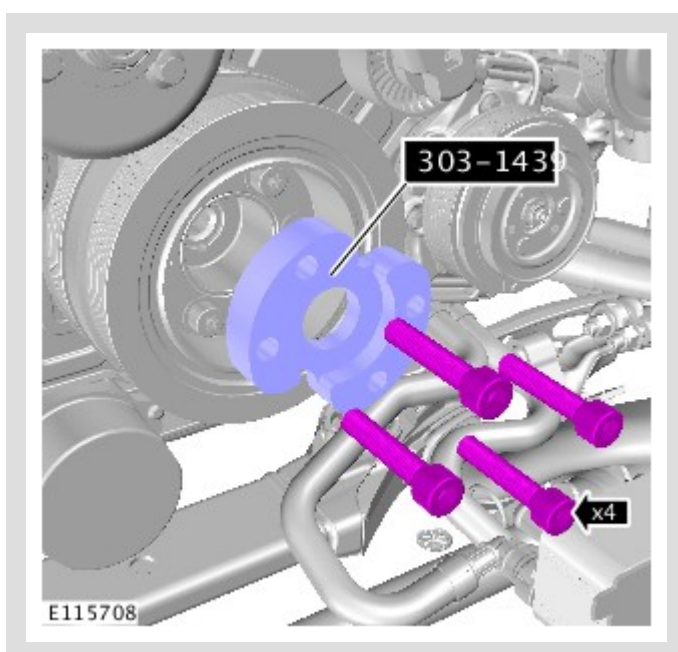


7.



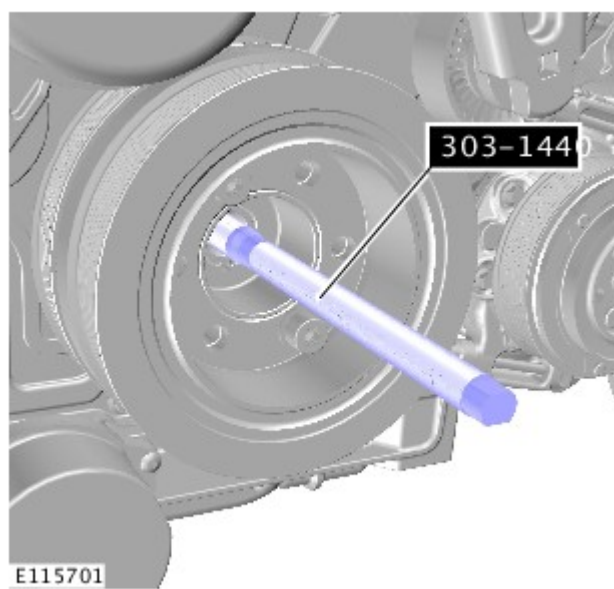
Remove the special tool.

8.



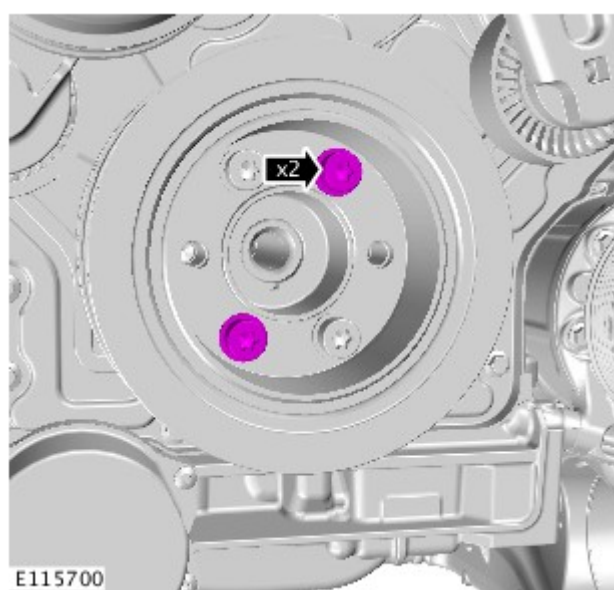
- Remove the special tool.
- *Special Tool(s):* [303-1439](#)

9.



Remove the special tool.

10.



 NOTE:

The original M10 bolts can be reused.

Clean the original crankshaft pulley bolts and apply loctite 270 to the threads before installation.

- **Torque: 65 Nm**

11.



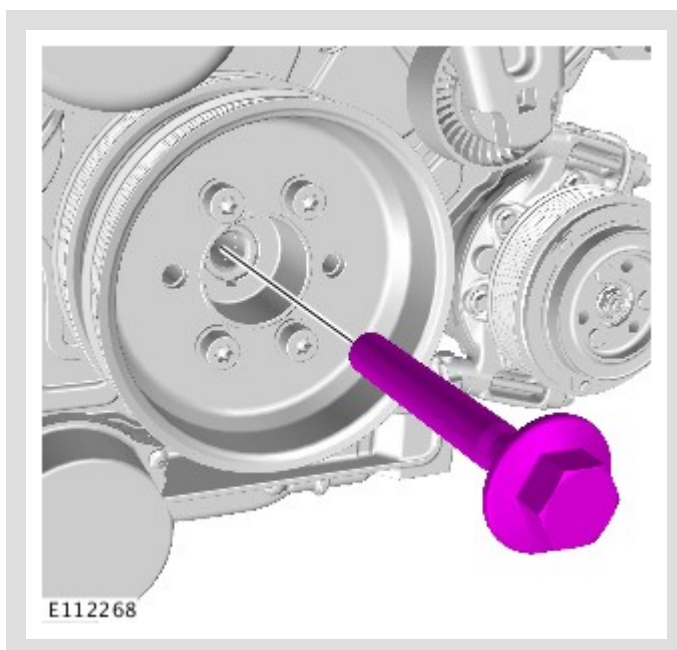
WARNING:

Make sure that a new bolt is installed.



CAUTION:

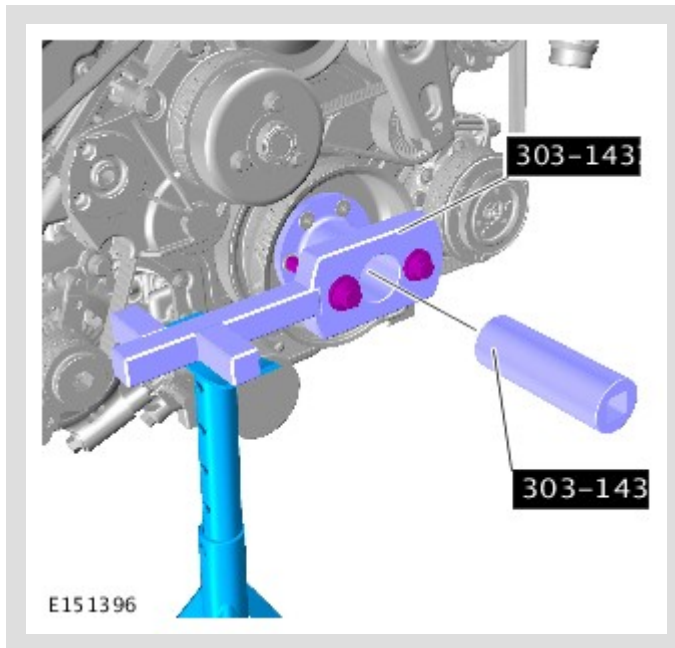
Tighten the component finger tight first.



Install the crankshaft pulley bolt.

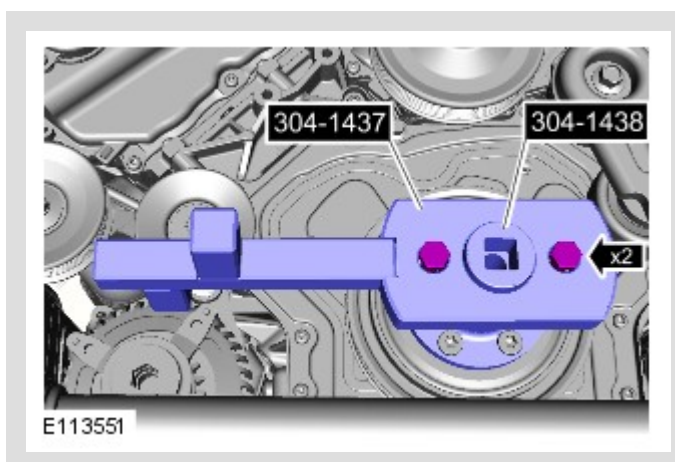
Renew Part: *Crankshaft pulley bolt* Quantity: *1* .

12.



- Install the special tools.
- *Special Tool(s):* 303-1437
- *Special Tool(s):* 303-1438

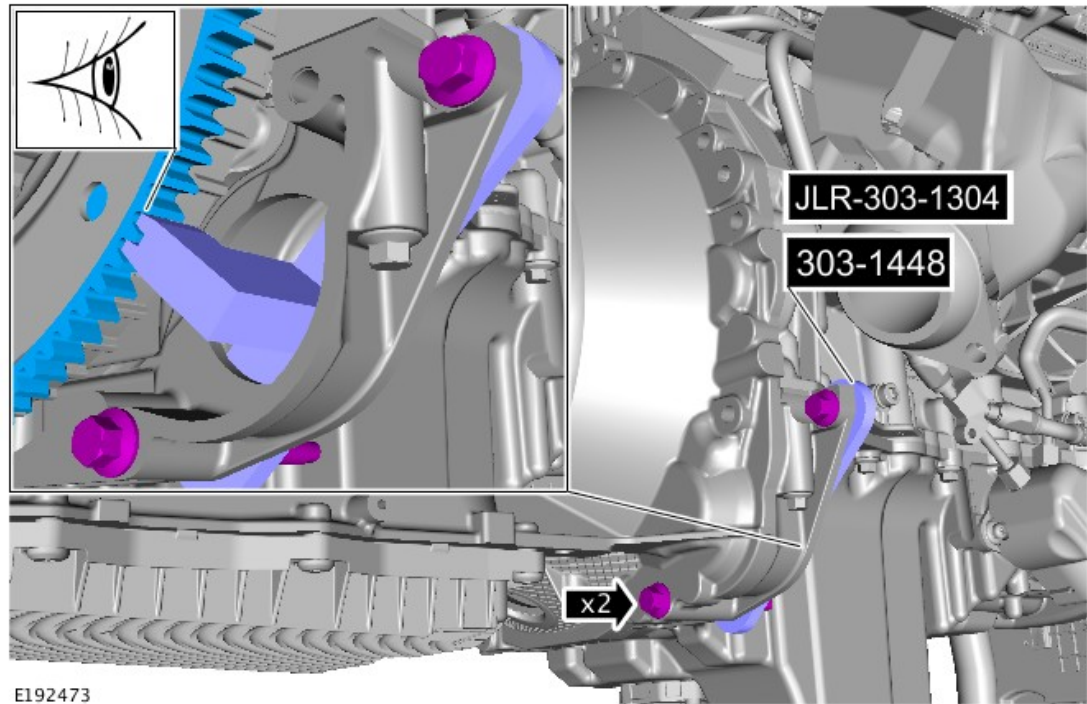
13.



Tighten the special tool bolts.

Torque: 25 Nm

14.

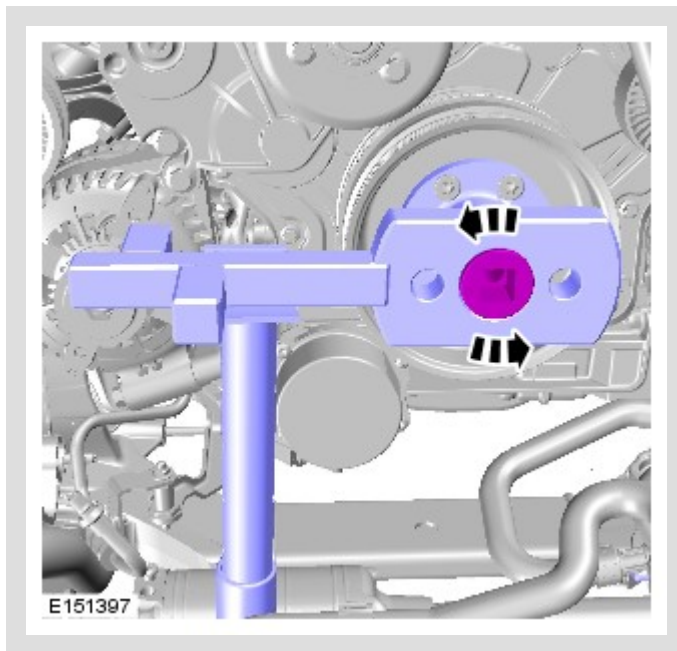


- Remove the special tool.
- Vehicles without Auto stop/start system.
Special Tool(s): [303-1448](#)
- Vehicles with Auto stop/start system.
Special Tool(s): [JLR-303-1304](#)

15.

△ NOTES:

- The use of a torque multiplier capable of 600Nm will be required.
- The illustration shows the tool positioning for LH threaded crankshaft damper bolts, RH threaded crankshaft bolts will require the tool position to be the opposite direction.



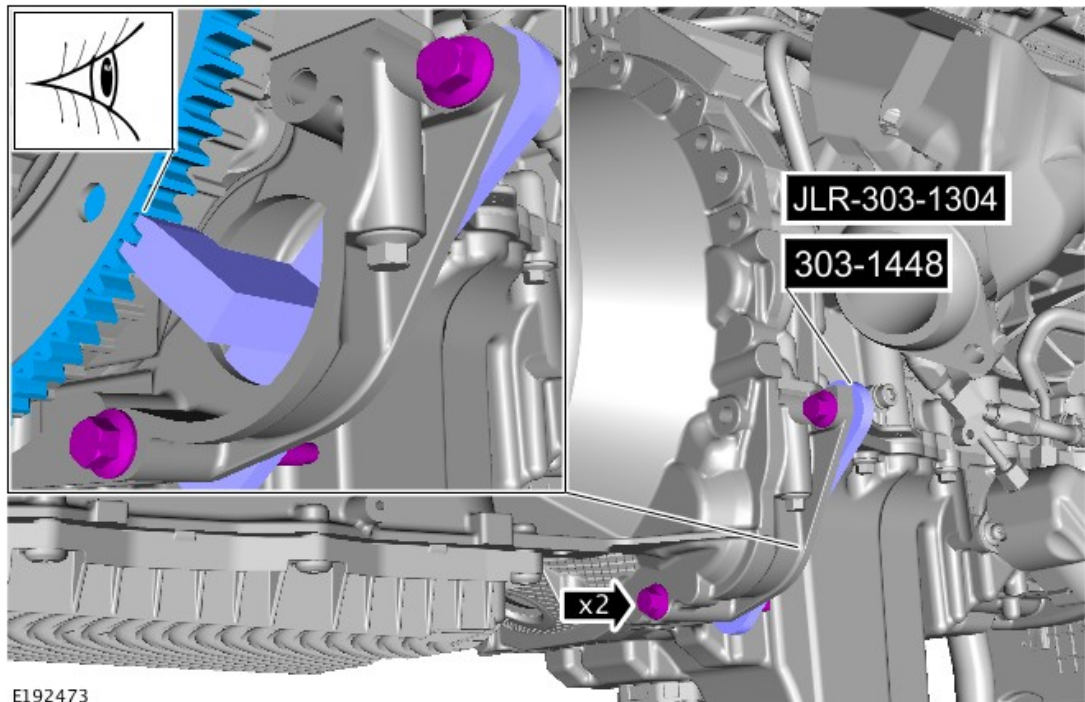
Tighten the crankshaft pulley bolt. Make sure that the socket is turned through 270 degrees not the torque wrench.

Torque:

Stage 1: **200 Nm**

Stage 2: **270°**

16.



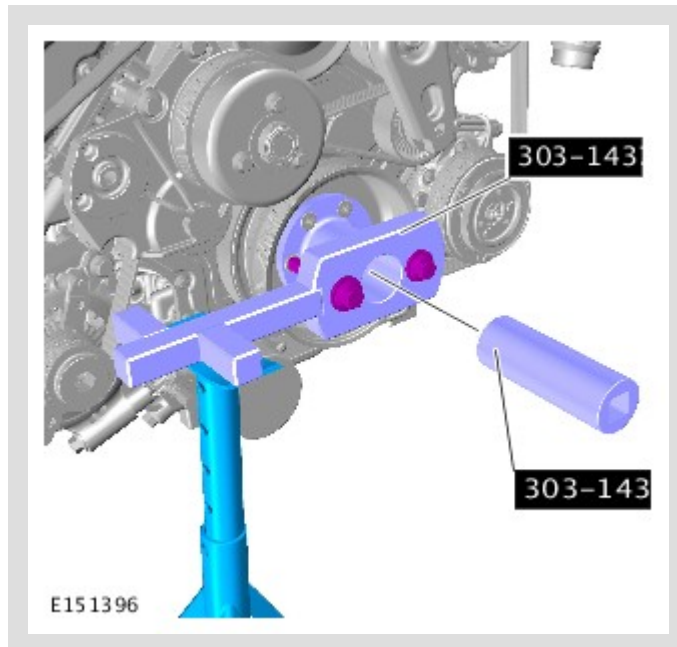
- Install the special tool.
- Vehicles without Auto stop/start system.

Special Tool(s): [303-1448](#)

- Vehicles with Auto stop/start system.

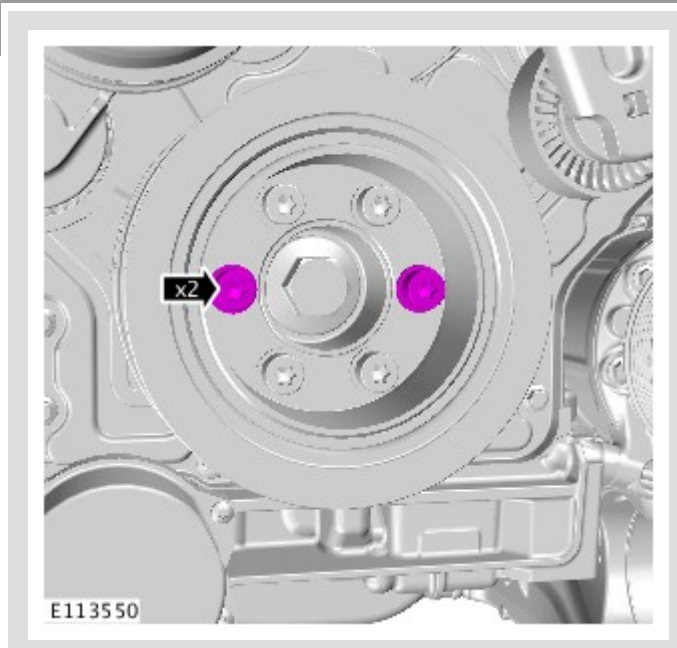
Special Tool(s): [JLR-303-1304](#)

17.



Remove the special tools.

18.



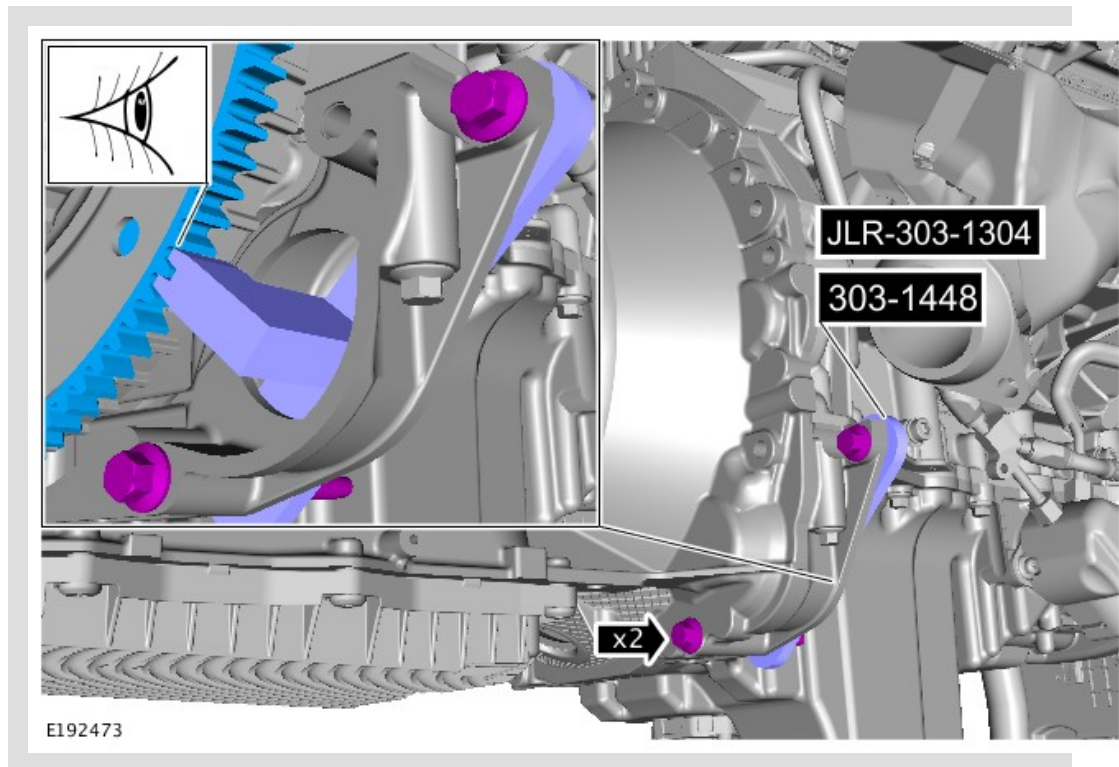
 **NOTE:**

The original M10 bolts can be reused.

Clean the original crankshaft pulley bolts and apply loctite 270 to the threads before installation.

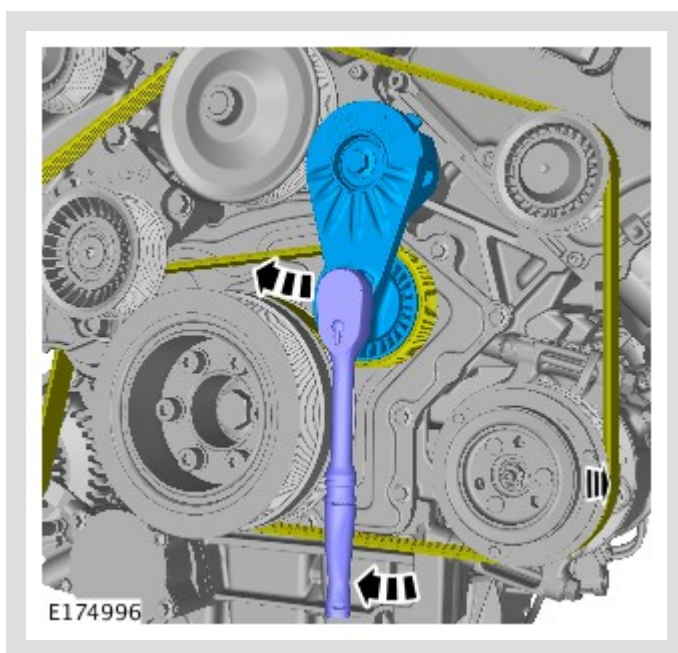
- **Torque: 65 Nm**

19.

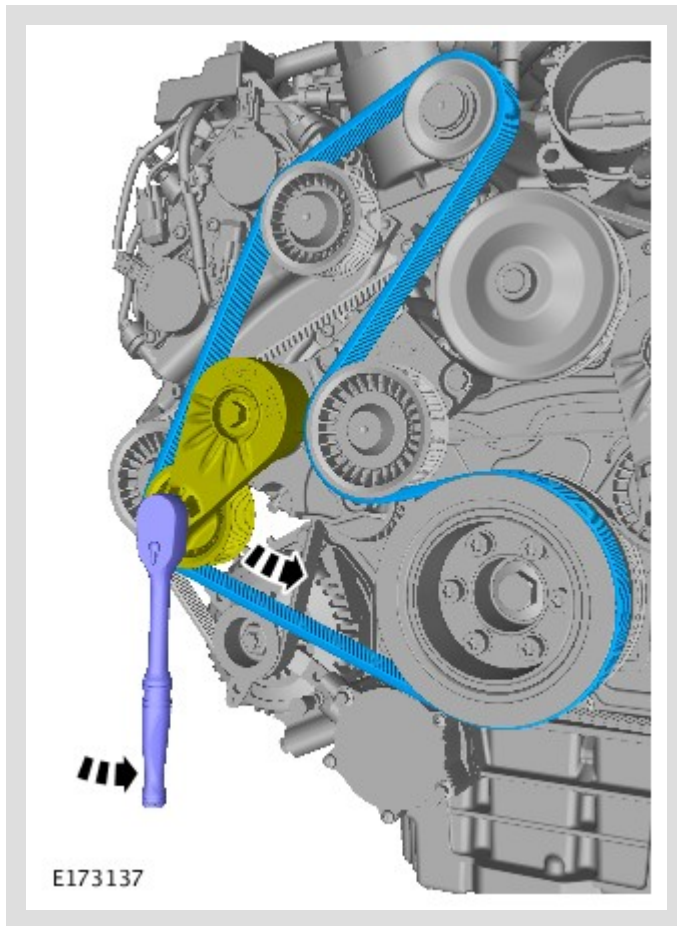


- Remove the special tool.
- Vehicles without Auto stop/start system.
Special Tool(s): [303-1448](#)
- Vehicles with Auto stop/start system.
Special Tool(s): [JLR-303-1304](#)

20.



Install the accessory drive belt.



Install the supercharger drive belt.

-
- | | |
|-----|---|
| 22. | <p>Install the cooling fan.</p> <p>Refer to: Cooling Fan Motor and Shroud (303-03 Engine Cooling - V6 S/C 3.0L Petrol, Removal and Installation).</p> |
|-----|---|
-
- | | |
|-----|---|
| 23. | <p>Install the starter motor.</p> <p>Refer to: Starter Motor (303-06 Starting System - V6 S/C 3.0L Petrol, Removal and Installation).</p> |
|-----|---|
-
- | | |
|-----|--|
| 24. | <p>Re-connect the startup battery ground cable.</p> <p>Refer to: Specifications (414-01 Battery, Mounting and Cables, Specifications).</p> |
|-----|--|